Contents

1 The Schrödinger Equation

1.1 Wave Functions

Example. An extremely common continuous distribution is the Gaussian distribution: $\rho(x) = Ae^{-\lambda(x-a)^2}$.

To normalize the distribution, we demand:

$$\int_{-\infty}^{\infty} Ae^{-\lambda(x-a)^2} dx = 1$$

Translating to be centered around the origin does not change the integral. Then let us substitute $\sqrt{\lambda}x = y$ we have

$$\frac{\sqrt{\lambda}}{A} = \int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$$

So $A = \sqrt{\frac{\lambda}{\pi}}$. Meanwhile, to find the expected value, make the substitution $(x-a) = y$, giving us

$$\langle x \rangle = A\int_{-\infty}^{\infty} ye^{-\lambda y^2} dy + aA\int_{-\infty}^{\infty} e^{-\lambda y^2} dy = -I_1 + I_1 + a = a$$

So we can expect the particle to be located at the middle of the gaussian distribution. Meanwhile,

$$\langle x^2 \rangle = A\int_{-\infty}^{\infty} y^2 e^{-\lambda y^2} dy + A\int_{-\infty}^{\infty} 2aye^{-\lambda y^2} dy + A\int_{-\infty}^{\infty} a^2 e^{-\lambda y^2} dy = \frac{1}{2\sqrt{\lambda}} + a^2$$

Therefore, the standard deviation σ_x is $\frac{1}{2\sqrt{\lambda}}$

Example. Here, we'll examine the time dependence of $\langle p \rangle$.

The derivative is:

$$\frac{d\langle p \rangle}{dt} = -i\hbar \int \frac{\partial}{\partial t}\left(\Psi^* \frac{\partial \Psi}{\partial x}\right) dx$$

using Schrödinger's equation along with the product rule. Expanding using the product rule and substituting in Schrödinger's equation gives us

$$\int \frac{\hbar^2}{2m}\left(-\frac{\partial^2\Psi}{\partial x^2}\frac{\partial\Psi}{\partial x} + \Psi^*\frac{\partial^3\Psi}{\partial x^3}\right) + V\Psi^*\frac{\partial\Psi}{\partial x} - \Psi^*\frac{\partial(V\Psi)}{\partial x} dx$$

Integrating by parts twice on the left term and applying the boundary conditions that $\frac{\partial\Psi}{\partial x}$ and $\frac{\partial\Psi^*}{\partial x}$ go to 0 at infinity cancels the two out. Expanding the right side, we have

$$\frac{\partial(V\Psi)}{\partial x} = \Psi\frac{\partial V}{\partial x} + V\frac{\partial\Psi}{\partial x}$$

so we are left with

$$\frac{d\langle p \rangle}{dt} = \int \Psi^*\left(-\frac{\partial V}{\partial x}\right)\Psi dx = \langle -\frac{\partial V}{\partial x} \rangle$$

This is Ehrenfest's theorem for Newton's second law.

Griffiths Problem 1.11 and 1.12 A classical approach to creating a wave function is by directly modeling the probability distribution of a classical particle using kinematics. Suppose a particle of mass m is in a classical harmonic oscillator of potential $V(x) = \frac{kx^2}{2}$ and has energy E. Let the fraction of time the particle spends at position x be

$$\rho(x)dx = \frac{dt}{T} = \frac{(dt/dx)dx}{T} = \frac{1}{v(x)T}dx$$

where T is the total period. Then the quantity $\rho(x) = \frac{1}{v(x)T}$ is a close analogy to $|\Psi|^2$. (Careful, T is the time it takes to travel from one side to the other, so it is half of the period)

Using conservation of energy, $v(x) = \sqrt{\frac{2(E-\frac{kx^2}{2})}{m}}$. Then we have $\rho(x) = \frac{1}{\pi\sqrt{(a^2-x^2)}}$. Integrating from a to -a, $\int_{-a}^{a} \frac{dx}{\pi\sqrt{a^2-x^2}} = \frac{1}{\pi}\left[\arcsin(\frac{x}{a})\right]_{-a}^{a} = 1$ so this "wave function" satisfies the normalization constraint. Meanwhile we have

$$\langle x \rangle = \int_{-a}^{a} \frac{x}{\pi\sqrt{a^2-x^2}}dx = \left[\frac{-\sqrt{a^2-x^2}}{\pi}\right]_{-a}^{a} = 0$$

$$\langle x^2 \rangle = \int_{-a}^{a} \frac{x^2}{\pi\sqrt{a^2-x^2}}dx = \frac{a^2}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2(\theta)d\theta = \frac{a^2}{2}$$

Thus, the standard deviation σ_x is $\frac{a}{\sqrt{2}}$. Meanwhile, if we were to do the same with momentum, we can write

$$\rho(p)dp = \frac{dt}{T} = \frac{(dt/dp)dp}{T} = \frac{dp}{T|\dot{p}|}$$

From conservation of energy, we have

$$\dot{p} = -\frac{dV}{dx} = -kx = \sqrt{2Ek - \omega^2 p^2}$$

Then the "momentum wave" is

$$\rho(p) = \frac{1}{(\pi/\omega)\sqrt{2kE - w^2p^2}} = \frac{1}{\pi\sqrt{2mE - p^2}} = \frac{1}{\pi\sqrt{p_0^2 - p^2}}$$

This is in the exact same form as the position wave, so we already know

$$\langle p \rangle = 0$$

$$\langle p^2 \rangle = \frac{p_0^2}{2}$$

Then using the standard deviations, we can see that

$$\sigma_x\sigma_p = \frac{p_0 x_0}{2}$$

This should be setting off alarm bells! From the Heisenberg Uncertainty Principle, we must have

$$\sigma_x\sigma_p \geq \frac{\hbar}{2}$$

but with our classical "wave," we can pick any arbitrarily small starting position. Clearly, this classical representation does not work.

1.2 Time Independent Schrödinger Equation

Example. Suppose the separation constant E is allowed to have an imaginary component. While an imaginary energy is physically impossible, there isn't immediately anything mathematically preventing this.

Let's consider a wavefunction whose energy has an imaginary component $E = E_0 + i\Gamma$ where Γ is strictly real. Then the wave function is $\Psi(x,t) = \psi(x)e^{-i\hbar(E_0+i\Gamma)/t}$. The normalization condition then gives us

$$1 = \int \Psi(x,t) = \int \psi^*\psi e^{2\Gamma t/\hbar} dx = e^{2\Gamma t/\hbar}$$

But in order for this to hold for all time t, this implies $\Gamma = 0$. Thus, as expected physically, the total energy must be strictly real.

Example. Suppose we have an imaginary solution $\psi(x)$. Then conjugating both sides of the time independent Schrödinger equation give us

$$-\frac{\hbar^2}{2m}\frac{d(\psi^*)^2}{dx^2} + V\psi^* = E\psi^*$$

since E (and consequently, V) must be strictly real. So the conjugate, real wave ψ^* is also a solution. But any linear combination of solutions is also a solution so both

$$\psi_1 = (\psi + \psi^*)$$

$$\psi_2 = i(\psi - \psi^*)$$

are also solutions. Then we can rewrite $\psi = \frac{1}{2}\psi_1 + \frac{1}{2i}\psi_2$. What this means is any time we have an imaginary solution to the Schrödinger solution, we can simply work with its real complex conjugate, and then if needed, convert back to the imaginary solution.

Example. If $V(x)$ is an even function we can see that if $\psi(x)$ is a solution, then

$$-\frac{\hbar^2}{2m}\frac{d\psi(-x)^2}{dx^2} + V(-x)\psi(-x) = E\psi^*$$

$$-\frac{\hbar^2}{2m}\frac{d\psi(-x)^2}{dx^2} + V(x)\psi(-x) = E\psi^*$$

$\psi(-x)$ is also a solution. But the linear combinations $\psi(x) + \psi(-x)$ and $\psi(x) - \psi(-x)$ are also solutions. $\psi(x) + \psi(-x)$ is even and $\psi(x) - \psi(-x)$. So in this case we can always just choose to work with even or odd wavefunctions.

1.3 Infinite Potential Well

Example. We'll examine the stationary states of the infinite potential well.
The spatial component is $\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$ so we have

$$\langle x \rangle = \frac{a}{2}$$

$$\langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2n^2\pi^2} \right)$$

We can easily compute $\langle p \rangle$ and $\langle p^2 \rangle$using Ehrenfest's theorem. Using $\langle p \rangle = m\frac{d\langle x \rangle}{dt}$ and $E_n = \frac{\langle p^2 \rangle}{2m}$ then we have

$$\langle p \rangle = 0$$

$$\langle p^2 \rangle = \frac{n^2\pi^2\hbar^2}{a^2}$$

This lines up with the uncertainty principle, as we have

$$\sigma_x \sigma_p = \sqrt{\left(\frac{a^2}{12} - \frac{a^2}{2n^2\pi^2} \right) \frac{n\pi\hbar}{a}} = \hbar \sqrt{\frac{n^2\pi^2}{2} - \frac{1}{2}}$$

which is minimized at the ground state $\sigma_x \sigma_p = \hbar\sqrt{\frac{\pi^2}{2} - \frac{1}{2}} \approx 0.5678\hbar \geq \frac{\hbar}{2}$.

Example. In principle, we can have any wave function as our starting condition as long as it's normalizable. Consider the wave function

$$\Psi(x, 0) = \begin{cases} Ax & 0 \leq x \leq \frac{a}{2} \\ A(a - x) & \frac{a}{2} \leq x \leq a \end{cases}$$

In order to be normalizable, we need $\int \Psi^2 dx = 2(\frac{A^2a^3}{24}) = 1$ which means $A = \frac{2\sqrt{3}}{a^{3/2}}$. Note that this function is even about the center of the well, so only the odd n components give us anything. Then using Fourier's trick:

$$c_n = \int_0^{\frac{a}{2}} Ax\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x \right) = \frac{4\sqrt{6}}{n^2\pi^2} \sin\left(\frac{n\pi}{2} \right)$$

So the wavefunction as a function of time is simply

$$\Psi(x, t) = \frac{4\sqrt{6}}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{(n-1)/2}}{n^2} \sin\left(\frac{n\pi x}{a} \right) e^{-iE_nt/\hbar}$$

The probability of finding energy E_1 is $\frac{96}{\pi^4} \approx 0.9855$ and the expectation of energy is $\langle H \rangle = \frac{96E_1}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{12}{\pi^2} E_1 \approx 1.216E_1$. Within the wave function, the ψ_1 terms dominate and this is reflected in the expectation of energy.